Sabit Hassan | Curriculum Vitae

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Founder & Chief Executive Officer @ Moonworks AI

EDUCATION

BSc in Computer Science *Carnegie Mellon University* 2014–2018
Minors: Mathematics and History
Thesis: Interactive Evaluation and Training of Classifiers under Limited Resources,
Advisors: Bhiksha Raj and Saquib Razak

PhD in Computer Science *University of Pittsburgh* Aug 2021–Jan 2025
Dissertation: Active Learning Frameworks for Safe, Inclusive, and Democratic Conversational AI
Committee: Malihe Alikhani (Co-Chair), Jacob Biehl (Co-Chair), Diane Litman, Xiaowei Jia, and Silvio Amir

HONORS AND AWARDS

Outstanding Paper Award	*Customizable NLP@EMNLP*	Nov 2024
Alexa Prize Taskbot Challenge Award	*Amazon*	Sep 2023
Alexa Prize Taskbot Challenge Grant	*Amazon*	Jan 2023
Kuzneski Innovation Cup Award	*The Big Idea Center*	Nov 2022
Outstanding Academic Achievement	*Carnegie Mellon University*	May 2018
University Honors	*Carnegie Mellon University*	May 2018
SCS College Honors	*Carnegie Mellon University*	May 2018
History Honors	*Phi Alpha Theta*	May 2018
Carnegie Mellon Dean's List	*Carnegie Mellon University*	Apr 2015–Apr2018
Best Design Award	*Carnegie Apps Hackathon*	Feb 2018
Best Technical Challenge Award	*Carnegie Apps Hackathon*	Feb 2017

RESEARCH

Doctoral Research *University of Pittsburgh* Aug 2021–Jan 2025
- Generative active learning, generative AI safety, dialogue systems, embodied agents
- Sign language generation, content moderation, contextual ASR, NLP in healthcare

Graduate Student Researcher *Amazon* Jan 2023–Sep 2023
- Led development of safety and retrieval systems for ISABEL, a multimodal task-oriented dialogue system at Alexa Taskbot Challenge 2

Researcher *Qatar Computing Research Institute* Oct 2019–Jun 2021
- Pre-trained language model (QARiB), dialect, offensive language, cross-lingual emotion detection
- Social Media Analytics Suite (ASAD), NLP processing tool (Farasa)

Research Associate *Carnegie Mellon University Qatar* May 2018–Sep 2021
- Co-organized MADAR shared task on Arabic dialect identification
- Characterized oracle hierarchy for small one-way finite automata

TEACHING

Teaching Fellow *University of Pittsburgh* Jan 2025–May 2025
- Course instructor for CS 0012: Introduction to Computing

Guest Lectures

- Active Learning for Democratizing Safety and Fairness in LLMs **Northeastern University,** Mar 22, 2024
- Causal inference in LLMs, **Northeastern University,** Mar 19, 2024
- Machine Translation, **University of Pittsburgh,** Nov 15, 2023

Graduate Teaching Assistant *University of Pittsburgh* Aug 2021-Aug 2024

- Intro to Natural Language Processing, Algorithms and Data Structures, Intro to Web Applications

Teaching Assistant *Carnegie Mellon University* Jan 2016-May 2018

- Freshman Edge Summer Program, Great Theoretical Ideas in CS, Parallel and Sequential Data Structures and Algorithms, Fundamentals of Programming and CS, Concepts of Mathematics

Summer Internship Mentor *Qatar Computing Research Institute* May 2020-Jul 2020

- Mentored interns on the use of machine learning in NLP tasks

SERVICES

- **Co-organizer,** MADAR Shared Task on Fine-Grained Arabic Dialect Identification, 2019
- **Program Committee/Reviewer:** ARR, ACL, EACL, EMNLP, IJCNLP-AACL, Journal of NLP, IEEE MM, Heilyon, ICARD
- **Member,** Student Academic Committee, Academic Review Board, Carnegie Mellon University Qatar

PUBLICATIONS [Google Scholar]

* denotes equal contribution

[1] Wang, Y., Hassan, P., Sadeka, S., Soliman, N., Abdullah, M M S., **Hassan, S**. *Moonworks Lunara Aesthetic II: An Image Variation Dataset.* 2026.

[2] Wang, Y., Abdullah, M M S., Hassan, P., **Hassan, S**. *Moonworks Lunara Aesthetic Dataset.* 2026.

[3] Alikhani, M. & **Hassan, S.,** *Hype and harm: Why we must ask harder questions about AI and its alignment with human values.* **BROOKINGS, 2025**

[4] **Hassan, S.,** Chung, H., Tan, Z., & Alikhani, M. *Coherence-driven Multimodal Safety Dialogue for Embodied Agents*. **AAMAS, 2025**

[5] **Hassan, S.,** Sicilia, A. & Alikhani, M. *An Active Learning Framework for Inclusive Generation by Large Language Models.* **COLING, 2025**

[6] Asano, Y., **Hassan, S.,** Sharma, P., Sicilia, A, Atwell, K., Litman, D., & Alikhani, M. *Contextual ASR Error Handling with LLMs Augmentation for Goal- Oriented Conversational AI.* **COLING, 2025**

[7] **Hassan, S.,** Sicilia, A., & Alikhani, M. *Active Learning for Robust and Representative LLM Generation in Safety-Critical Scenarios.* **Customizable NLP @ EMNLP 2024 [Outstanding Paper Award]**

[8] Sicilia, A., Asano, Y., Atwell, K., Cheng, Q., Gupta, D., **Hassan, S.** Inan, M., Nwogu, J., Sharma, P., Alikhani, M. *ISABEL: An Inclusive and Collaborative Task-Oriented Dialogue System.* **Amazon Science, 2023. [Alexa Prize Taskbot Challenge Award]**

[9] **Hassan, S**. & Alikhani, M. *DisCGen: A Framework for Discourse-Aware Counterspeech Generation.* **IJCNLP-AACL, 2023**

[10] **Hassan, S**. & Alikhani, M. *D-CALM: A Dynamic Clustering-based Active Learning Approach for Mitigating Bias.* **ACL FINDINGS, 2023**

[11] Wang, Y., Donovan, H.A.S., **Hassan, S**., Alikhani, M**.** *MedNgage: A Dataset for Understanding Engagement in Patient-Nurse Conversations.* **ACL FINDINGS, 2023**

[12] Ye, M., Sikka, K., Atwell, K., **Hassan, S.,** Divakaran, A., Alikhani, M. *Multilingual Content Moderation and Challenges: A Case Study of Reddit.* **EACL, 2023**

[13] Mubarak, H., **Hassan, S.,** & Chowdhury, S. *Emojis as Anchors to Detect Arabic Offensive Language and Hate Speech.* **NLE, 2023**.

[14] Atwell, K.*, **Hassan, S.*,** Alikhani, M. *APPDIA: A Discourse-aware Transformer-based Style Transfer Model for Offensive Social Media Conversations.* **COLING, 2022**

[15] Inan, M.*, Yang, Z.*, **Hassan, S.*,** Quandt, L., & Alikhani, M. *Modeling Intensification for Signed Language Generation: A Computational Approach.* **ACL FINDINGS, 2022**

[16] **Hassan, S.,** Shaar, S., & Darwish, K. *Cross-lingual Emotion Detection.* **LREC, 2022**

[17] Mubarak, H., **Hassan, S.**, Chowdhury, S. A., & Alam, F. *Analyzing Arabic Tweets About COVID-19 Vaccination*. **LREC, 2022**.

[18] **Hassan, S.,** Mubarak, H., Abdelali, A., & Darwish, K. *ASAD: Arabic Social Media Analytics and unDerstanding.* **EACL 2021**

[19] Mubarak, H., **Hassan, S.,** & Abdealali, A. *Adult Content Detection on Arabic Twitter: Analysis and Experiments.* **WANLP 2021**

[20] Abdelali, A., Mubarak, H., Samih, Y., **Hassan, S.,** & Darwish, K. *QADI: Arabic Dialect Identification in the Wild.* **WANLP 2021**

[21] Mubarak, H., & **Hassan, S.** *Analyzing Arabic Tweets in the Early Days of Coronavirus (COVID-19) Pandemic.* **LOUHI, 2021**

[22] Mubarak, H., Abdelali, A., **Hassan, S**., & Darwish, K. *Spam Detection on Arabic Twitter.* **SocInfo, 2020**

[23] **Hassan, S.,** Samih, Y., Mubarak, H., & Abdelali, A. *ALT at SemEval-2020 task 12: Arabic and English offensive language identification in social media.* **SemEval, 2020**

[24] **Hassan, S.,** Samih, Y., Mubarak, H., Abdelali, A., Rashed, A., & Chowdhury, S. A. *ALT at OSACT Shared Task on Offensive Language Detection.* **OSACT, 2020 [Ranked 1st in subtask A]**

[25] Mubarak, H., **Hassan, S.,** & Abdelali, A. *Constructing a bilingual corpus of parallel tweets.* **BUCC, 2020**

[26] Bouamor, H., **Hassan, S.,** & Habash, N. *The MADAR Shared task on Arabic Fine-Grained Dialect Identification.* **WANLP, 2019**. **[co-organizer]**

[27] Anabtawi, M., **Hassan, S.,** Kapoutsis, C., & Zakzok, M. *An Oracle Hierarchy for Small One-way Finite Automata.* **LATA, 2019**

[28] **Hassan, S.,** Shaar, S., Raj, B., & Razak, S. *Interactive Evaluation of Classifiers Under Limited Resources.* **IEEE ICMLA, 2018**

[29] Abdelali, A., **Hassan, S.,** Mubarak, H., Darwish, K., & Samih, Y. *Pre-Training BERT on Arabic Tweets: Practical Considerations.* **arXiv preprint arxiv:2102.10684 (2021)**

[30] **Hassan, S.***, Atwell, KJ.*, Alikhani, M. *Studying the Effect of Moderator Biases on the Diversity of Online Discussions: A Computational Cross-linguistic Study.* **CogSci, 2022**

[31] Inan, M.*, Zhong, Y.*, **Hassan, S.***, Quandt, L., Alikhani, M. *Learning cognitive and linguistic prosodic categories for automatic cross-lingual sign language understanding.* **CogSci, 2022**